MAHMOUD ABD EL-GHAFFAR

Computer Engineer | AI Automation & Web Development Specialist

Website Design & UI/UX | AI-Assisted Workflows | Make, Zapier & n8n | Live Render Deployments | APIs & CRMs

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PROFESSIONAL SUMMARY

Accomplished Computer Engineer and AI Automation Specialist combining deep expertise in full-stack website design and development with robust systems integration. Experienced in designing visually compelling, mobile-first websites and web portals utilizing modern AI design workflows (Google Antigravity, Claude Code, rapid UI prototyping) to accelerate delivery. Proven portfolio of live concept web design mockups and interactive prototypes deployed on Render (including dental clinic mockups like Smile Inn, sovereign institutional portal concepts like FOE Procurement Portal, task assistant apps, and personal portfolio). Co-built 30+ native integration nodes and automated 30+ production pipelines connecting CRMs (HubSpot, GoHighLevel, Airtable), forms (Typeform, Tally, Fillout), messaging platforms (WhatsApp Cloud API/WAPA, Instagram, Telegram), and databases. Comfortable working under strict NDA and proprietary data standards.

CORE TECHNICAL COMPETENCIES

- **Website Design & Front-End UI/UX:** Responsive Web Design (Mobile/Tablet/Desktop), UI/UX Layouts, Visual Hierarchy, Modern CSS3 (Grid, Flexbox, Animations), HTML5, JavaScript (ES6+), React / Vite, Glassmorphism.
- **AI-Powered Design & Engineering Workflows:** Google Antigravity, Claude Code, AI wireframing, prompt-to-UI component generation, rapid design iteration, asset optimization, design-to-code translation.
- **Automation Platforms & Orchestration:** Make (Integromat), Zapier, n8n, AI MicroMind (30+ Native Core Nodes), Webhooks, Cron Schedulers, REST APIs, OAuth2.
- **Social & Messaging Automation:** Meta WhatsApp Cloud API & WAPA, Instagram DM & Comment Funnels, Facebook Messenger Bots, Telegram Bot API, Slack, Gmail, ClickUp.
- **AI & Multi-Agent Frameworks:** LangChain (Expert), LangGraph (Advanced), Model Context Protocol (MCP), Pinecone Vector DB, OpenAI Assistants, Vapi Voice AI.
- **Cloud Infrastructure & Full-Stack:** Node.js, TypeScript, Python, Render Web Services, Docker, AWS, DigitalOcean Linux VPS, PostgreSQL, SQL Server, GitHub CI/CD.

PROFESSIONAL EXPERIENCE

AI, Web & Automation Engineer — AI MicroMind

2024 – Present

- **AI-Assisted Web Design & UI Development:** Designed and built responsive web applications, micro-apps, and custom UI components utilizing modern AI developer tools (Google Antigravity, Claude Code), accelerating prototyping, styling, and live deployment.
- **Native Integration Nodes:** Built and co-built 30+ native integration nodes directly inside AI MicroMind Core (Node.js runtime) for WAPA (WhatsApp API), Telegram, Gmail, Weather, MCP Client, Pinecone, Webhooks, Schedulers, and PostgreSQL Audit Loggers using AI coding tools.
- **Multi-Step Business Automations:** Designed automated workflows across Zapier, Make, and n8n connecting CRMs (HubSpot, GoHighLevel, Airtable), web forms (Typeform, Tally, Fillout), and social channels (WhatsApp, Instagram, Telegram), eliminating manual data entry.
- **Conversational BI & Report Bot Framework:** Architected the AI MicroMind App featuring the Report Bot Framework, enabling non-technical users to query on-premise ERPs (Oracle, SAP, Microsoft Dynamics, Odoo, SQL) and cloud databases via natural language.
- **Proactive Error Handling & 99.9% Uptime:** Implemented robust fallback triggers, alert notifications, and webhook retry mechanisms, ensuring zero data loss and fault-tolerant production execution.
- **Multi-Agent Graphs & MCP:** Constructed stateful multi-agent graph workflows using LangChain and LangGraph with Model Context Protocol (MCP) server dispatchers for secure backend script execution.

PROFESSIONAL EXPERIENCE (CONTINUED)

Front-End & Automation Engineer — Operatives (IT Services)

Jun 2021 – Jan 2022

- **Web Portals & Workflow Synchronization:** Developed internal web portal interfaces, automated dashboards, and data synchronization pipelines, eliminating repetitive manual bottlenecks and reducing data-entry errors by over 70%.
- **Integration Reliability:** Maintained multi-system integrations across internal service catalogs, ticketing systems, and database stores with automated error logging and alerting.

Operation Engineer & Admin — Operatives (IT Services)

Jan 2020 – Jun 2021

- **IT Infrastructure & Operations:** Administered IT infrastructure operations, monitored workflow execution logs, enforced strict SLA compliance, and resolved technical service tickets with prompt escalation and root-cause analysis.
- **System Health & Maintenance:** Conducted regular preventative maintenance, scheduled automated backup routines, and drafted operational runbooks for technical staff.

KEY TECHNICAL ARCHITECTURE PROJECTS

- **Smile Inn Dental Clinic Website (Live Concept Mockup — Render):** Designed a modern, responsive clinic website concept mockup featuring calming color psychology, interactive before/after treatment sliders, doctor profiles, and appointment booking funnels.
- **FOE Sovereign Procurement Directorate Portal (Concept Prototype — Render):** Conceptual digital web portal and dashboard mockup designed for sovereign procurement operations with executive branding, responsive data tables, multi-role navigation, and bilingual Arabic/English layout.
- **Omnichannel Social & Messaging Automation:** Multi-platform social funnels connecting WhatsApp Cloud API / WAPA, Instagram DM comment-to-lead triggers, Facebook Messenger support bots, and Telegram channels with HubSpot & GoHighLevel.
- **Advanced Forms & Lead Capture Pipelines:** Multi-step conditional form routing across Typeform, Tally, Fillout, and Jotform with automated data enrichment, calendar booking, and instant Slack/WhatsApp sales alerts.
- **AI MicroMind App & Report Bot Framework:** Conversational BI reporting engine translating natural language prompts into live SQL/OData queries across Oracle, SAP, Dynamics 365, Odoo, and SQL databases.
- **Smart Task Assistant & Merna Invitation Web Apps (Live Prototypes — Render):** AI-powered task management app prototype with sleek dark-mode glassmorphism alongside a bespoke mobile-first animated digital invitation web mockup.

EDUCATION

B.Sc. in Computer Engineering — Faculty of Engineering Shoubra, Benha University

Benha, Egypt

- **Specialization:** Computer Systems Architecture, Database Management Systems, Software Engineering, Web Technologies & AI Systems.

PROFESSIONAL STANDARDS & WORKING STYLE

- **Full-Cycle Web & Automation Delivery:** Capable of taking a digital product from concept, wireframe, and visual styling to responsive code, cloud deployment on Render, and automated backend integrations.
- **AI-Enhanced Speed & Quality:** Utilizes AI tools as a force multiplier to accelerate design iterations, optimize assets, and deliver production-ready assets with high fidelity.
- **Autonomous Remote Execution & NDA:** Self-driven remote professional who communicates clearly in plain language, solves breaks independently, and strictly respects client confidentiality (NDA).